# Drone Detection And Drone Type Classification Using Deep Learning Technique


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***Abstract*— The demand and usage of drones are increasing day by day, and with this increased popularity, new types of drones are coming into the market for various purposes. However, due to the inflation in the usage of drones, their security has become a major concern. As a result, it is required for drone users to be well aware of drones surrounding them. To address this issue, a complete drone detection and classification scheme has been proposed that depends on deep learning. The scheme involves two parts: drone detection using YOLOV2 and drone classification using Convolutional Neural Networks (CNN). In the existing method, Support Vector Machine (SVM) is used to categorize the drone type as Tricopter or Quadcopter. The results of the proposed deep learning and Artificial Intelligence technology executed in MATLAB are promising. It's worth noting that the detection and classification of drones using deep learning algorithms is an active area of research. There are several other approaches that have been proposed in recent years, such as using radio frequency compressed signals to detect and classify UAVs 1. Another study proposed a hybrid deep learning model that combines LSTM and Bi-GRU classifiers to provide enhanced object detection in drone imagery for search and rescue operations 2. These approaches show great potential for improving the accuracy and efficiency of drone detection and classification.**



***Keywords—Drone detection, classification, YOLO, CNN, SVM Deep learning***


## 1.INTRODUCTION

A drone or Unmanned Aerial Vehicles (UAVs), in a process of summarize, is an unmanned airplane. Fundamentally, an Unmanned Aerial Vehicles (UAVs) is assumed as a airborne robot. The aircraft possibly controlled using controller or can take off separately thru programming-handled aeronautical planes with in-build structures operating at the side of onboard sensors and GPS .In the near previous years, Drones have been mostly used in the military, wherein those were used for anti-plane target practice at first, intelligence has been increased by then after which, more polemically, as arms systems. Drones are presently extensively utilized in a huge variety of citizen roles starting from search, rescue, scrutiny, traffic chasing, conditions checking, and also for industrial purposes, as well as videography, cultivation, or also for few small transportation purposes. The location of drones as such information is provided to army to provide data. The first actual aircraft with a reusable type radio manipulation mechanism was designed in the 30s and was it labels like a base version for all new advancements nowadays days global.

Later, the military drones have been developed with traditional sensors and digital camera gadgets and now they have been fixed internal missiles too. As technology is increasing rapidly different types of drones are also increasing. Few are used for navy programs but others are locating the capability function in many big organizations. And mostly due to advantages of drones many industries like Amazon are trying to bring their own drone for delivery of products purpose[9]. One extra thrilling idea is offered by Facebook as they may be wondering to increase a




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few large drones which can convey the signal to remote locations for direct internet access[1].

There are more chances for UAV-IOT networks providing airborne IoT services to have a few security issues. Several safety exposures like chances of a takeover, blocking of a UAV in flight, inactivating the drones or planes, and many others. They have been marked as the famous UAVs accessible within the marketplace[2]. Since the susceptibilities are public to be there, this type of UAV has an extreme chance of ill-use. Cyberattacks on commercial services have become common nowadays. As UAVs have become a vital part of the commercial staff, it is more important to have high security levels to undergo commercial and non-commercial operations.

## II.EXISTING METHOD

A synthetic neural community is an adaptive device that learns by using linked nodes or neurons in a layered structure that mimics a human brain. A neural network may be trained to recognize styles, categorize statistics, and predict future occurrences by analyzing data[6].

The entire structure is broken down into levels of abstraction by a neural community It, like the human brain, can be taught to recognize styles in speech or images through the use of many instances.

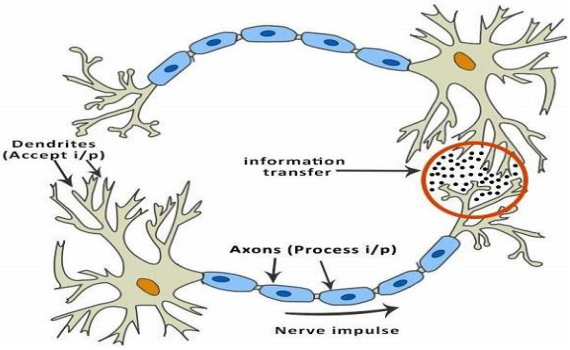



Fig 1 Human Neural Network Structure

Its behavior is determined by how its male and female elements are linked, as well as the strength, or weights, of those connections. During training, these weights are robotically altered in accordance with a specific learning rule until the synthetic neural network is capable of performing the desired task[7].

### *2.1 Artificial Neural Networks (ANN) Structure*

The working of ANN is totally based on the model of the human brain with the help of proper connection as in our brain like dendrites and neurons here we use silicon and wires. Our human brain consists of 86 billion nerve cells known as neurons. They are again connected to different cells with the help of Dendrites. Dendrites are used to respond for stimuli from outside world with the help of sense organs. This stimulation creates an electric impulse near dendrites to make communication faster in our brain. Only then a neuron can send info to a different neuron and helps for humans to respond immediately for actions occurring with them[3].

Similar to human brain ANN has multiple nodes in it. The neurons are connected with dendrites and hence with every other neuron in brain for communication. The nodes in ANN similar to neurons in brain collect records given to it and perform some easy operations. And the output is passed on to other nodes as well. The result of ANN is known as node value. Each link have it's own weight corresponding to it[4,5]. ANNs are capable of monitoring it's area by manipulating weight values. The following instance indicates a simple ANN.This ANN has nodes similar to neurons and these are connected in network format[5].

Each node is connected with other node with it's numerical weights, and those weights are saved in neural network and will be updating upto date and it is known as learning[9]. An ANN have three layers of nodes in it, they are input layer, output layer and hidden layer.

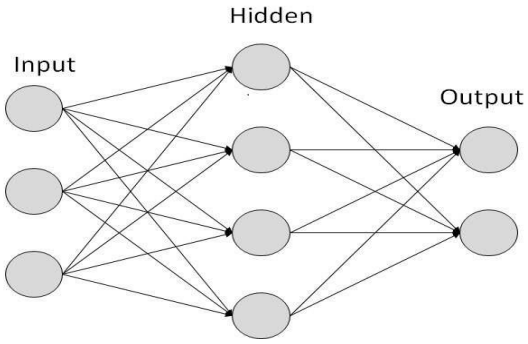



Fig 2 Layers of ANN

### *2.2 Working of an ANN*

In the above diagram we can see that every arrow represents a connection between two nodes and they shows the interconnection of nodes in ANN structure. Based on the weights provided to each node communication between nodes occurs. If those connections provide us with required output then there is no need to alter weights. However, if the network generates an unsatisfied output then the system changes weights if required[8].

### *2.3 Advantages of Artificial Neural Networks (ANN):*

- A network slowly might degrade and it will not affect the network much.



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- Artificial neural networks analyze every event and make decisions based on requirement and undergo changes.
- Artificial neural networks has a capability to undergo multiple processes at a time.

*2.4 Support Vector Machines (SVM) in ANN*

SVM is a type of Support vector machines existing learning method among the best available supervised algorithms present. But consistently, they're employed in class troubles. SVMs have been introduced in the year of 1960s but they have come into limelight in 1990. SVMs is the best way of implementation in comparison to any other machine learning algorithms. Recently they have come into extensive usage due their ability to deal with a few non-stop and express literals.

*2.4.1 Working of SVM*

An SVM version is a multidimensional representation of various data in a hyperplane. The hyperplane is built iteratively with the help of SVM in order to reduce any errors that may occur. The goal of SVM is to separate multi-dimensional datasets into various data sets to define a maximum marginal hyper aircraft (MMH).

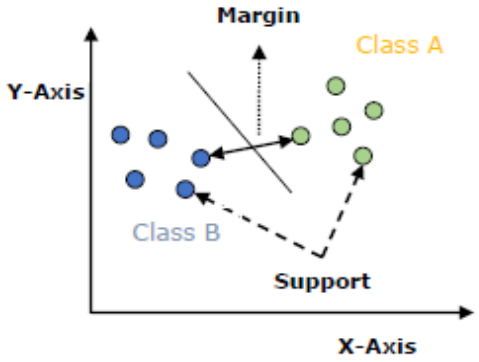



Fig 3 SVM model graph

The given below are vital features in SVM

- Support Vectors: Support vectors are data points that are closer to the hyperplane, and they are used to characterize the separation line.
- Hyper plane: A decision plane or space used to separate a collection of things into multiple classes.
- Margin: The distance between two lines formed by support vectors of various types. A large margin is considered a healthy margin, whereas a tiny margin is considered a severe margin.

To identify a maximum marginal hyper plane (MMH), SVM is used to partition datasets into distinct groups, as shown in the methods below.

- Firstly, it will create a hyperplane frequently that can separate the modules in a better way.
- After, it can easily select the hyperplane that divides the classes/Modules correctly.

*2.4.2 Training a support vector machine (SVM) has two stages*

- Transform the predictors (or input data) to a higher-dimensional space. It is sufficient to simply state the kernel for this step and the data is in no way unambiguously converted to the feature space. This method is normally referred to as the kernel trick.

- Solve quadratic optimization trouble to match an most excellent hyper aircraft to categorize the converted functions into classes. The amount of transformed features is determined with the useful resource of the usage of the amount of support vectors.

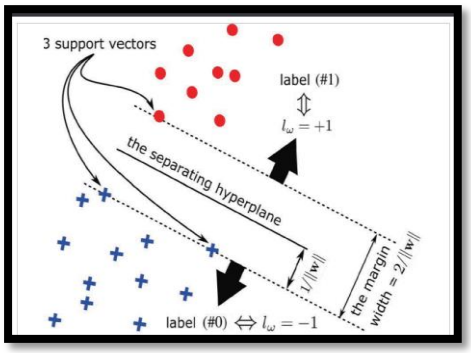



Fig 4 SVM graph of data separation

The "one versus all" and "one against one" SVM-based approaches are used to conduct the multi-class classification.



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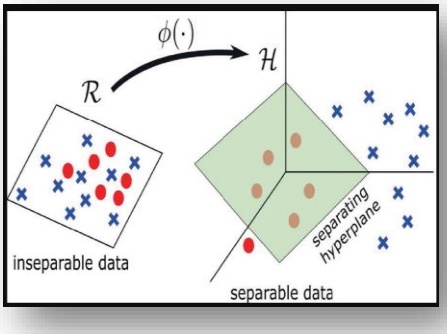



Fig 5 Hyperplane SVM graph

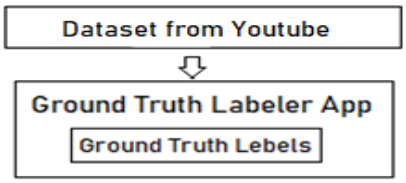
SVM Classifier "One Against One" In addition, the suggested strategy employs the "one versus one" SVM classification algorithm. For each pair of classes, k(k 1)/ 2 classifiers should be built, and the max-win method should be used. If sqn(wjl)T(xi) bjl determines that xi belongs to the j-th class, the vote for the j-th class is raised by one, and the vote for the l-th class is increased by one. Finally, it is projected that the training data vector xi will belong to the class with the most votes.

**Disadvantages:**

- Finding the ideal kernel function isn't always simple.
- Large datasets require a long training period.
- The final model, variable weights, and individual influence are difficult to comprehend and describe.
- Why Because the most recent version isn't always simple to detect, we can't make minor changes to the model, making it harder to keep our enterprise logic contained.
- The hyperplane parameters for Support Vector Machines are Cost -C and gamma. Improvements in these hyper-parameters are not simple to come by. It's difficult to predict their influence.

## III.PROPOSED METHOD

Here, we are going to speak about the Yolo and CNN network. Below figures show the model diagram of our proposed method.



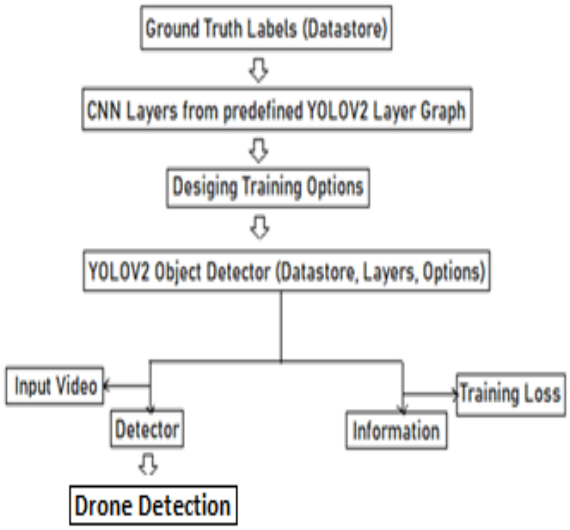




Fig 6 Block Diagram of Drone Detection using YOLO algorithm (Deep Learning)

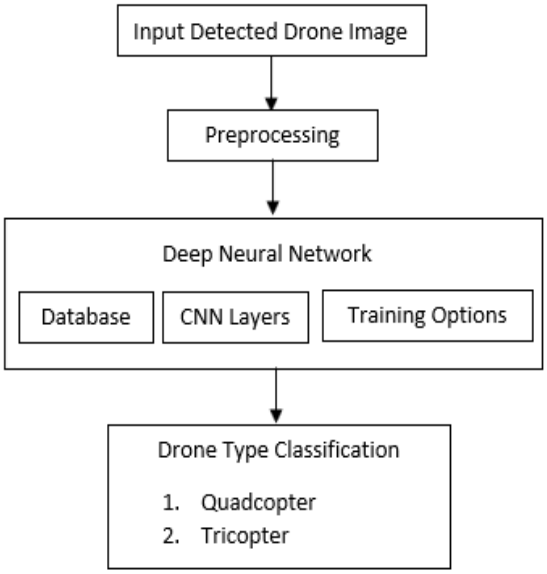



Fig 7 Drone Type Classification Using CNN (Deep Learning)

### *2.1 Layers of YOLO Model*

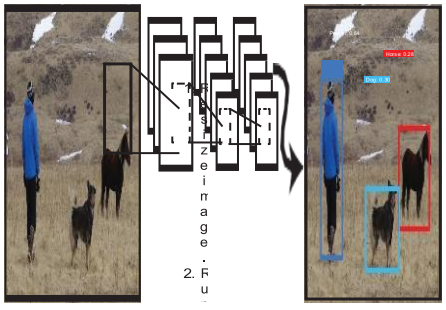


Fig 8 The YOLO Detection System



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In YOLOv2 each block of convolution have a Batch Norm normalization after which Leaky Relu activation except for the last block of convolution. The provided snapshot is divided into an SXS grid by YOLO. Each grid cell can have mostly one item. For example, if we consider a picture of a person detected in a yellow grid with a blue dot at the center, then the yellow grid cell tries to find the "person" object whose center (the blue dot) is within the grid cell. Boundary boxes are predicted for each grid cell. In this case, the yellow grid predicts two bounding boxes to determine which character it is. However, the only-item rule limits how near detected gadgets may be.

For each and every grid cell,

- It forecasts K boundary bins and every box has one field self-belief value,
- It predicts one object simplest regardless of the variety of bins K,
- •It predicts Y conditional-elegance-possibilities.

The boundary containers contain field self-assurance rating. The confidence rating displays how plausible the box involves an item (objectless) and by means of what correct is the frontier container. With image width and peak/top, we control the bounding box width and height/top. The offsets to the comparable mobile are x and y. As a result, x, y, w, and h all have values between 0 and 1. There are 20 conditional-elegance chances in each cell. The possibility that the seen object belongs to a certain magnificence is the conditional -elegance chance (1 chance for each class for every cell).The magnificence self- assurance rating for each prediction field is computed as:

Magnificence self-belief score = container self-assurance score * conditional magnificence probability.

It assesses the level of certainty in both categorization and localization. If necessary, we may easily blend those value(scoring) and probability terms. The references for the mathematical definition are listed below.

YOLO predicts more than one bounding bins according to grid mobile. To measure the loss for the high quality we simply need at least one of them in every cell grid to be accountable for the object. As a result, the one with the best IoU (intersection over union) with the ground fact must be chosen.

This method results in the specialization of many of the bounding container predictions. Each prediction reaches its best at predicting the size and factor ratio.It uses its overall errors between the predictions and the ground truth labeler model to detect the loss. The loss characteristic consists of:

- Category loss.
- The localization loss
- And the self-assurance loss

*3.1 CONVOLUTIONAL NEURAL NETWORKS (CNNS)*

CNNs uses the Bacak-propagation method to always update the weights in every adjoining layer. Therefore, CNNs process enables to us to measure the load update. We must have to run this method numerous times till convergence completely occurs. Some-other things like the usage of Fast Fourier Transform algorithms to clear out the enter statistics or using max-pooling to subsampling are associated with either the convolution manner or sub-sampling process. Mostly they are primarily created at the classical techniques in signal processing.

*3.2 ADVANTAGES AND APPLICATIONS*

*3.2.1 Advantages:*

i. YOLO forecasts the bounding box coordinates and class probabilities for those boxes based on the complete picture in a single occurrence.

ii. The most significant advantage of using YOLO is its incredible speed — it processes 45 frames per second.

iii. YOLO also understands generic item illustration. The most significant advantage of CNN over its predecessors is that it detects vital processes without the need for human intervention. Given several photographs of cats and puppies, for example, it learns unique functions for each elegance by utilizing itself.

iv.CNN maintains its computational efficiency. It makes use of distinct loop and pooling procedures, as well as parameter sharing. This allows CNN models to operate on any device, which benefits the ruling class in general attractive.

*3.2.2 Applications:*

1. It is used in Military for updating few important data for security purposes.
2.It can be developed for the purpose of Delivery.
3. If required in any Emergency Rescue schemes.
4.It also be used in spraying water for Crops.
5. It may be used in Outer Space.
6.Also, for Wildlife and Historical Conservation purpose by the government for wildlife protection.
7.This can also be used in some fields like Medicine, Photography, Mapping and surveying and Crop monitoring.



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## IV.RESULTS



Fig 9 Input Video

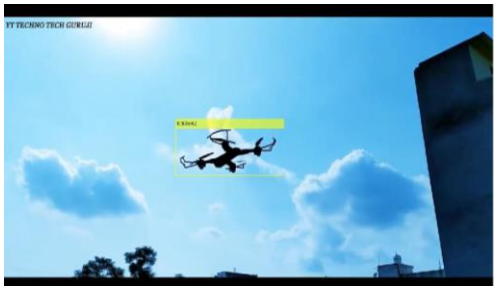
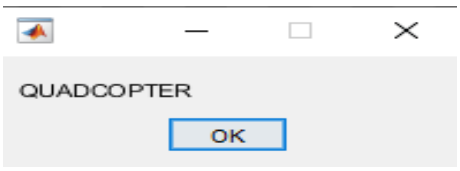


Fig 10 Drone Detection Video



Fig 11 Detected Drone

### *4.1 Training Progress*




### *4.2 Classified Drone Type*




Fig 12 Classification

## V.CONCLUSION

Finally, we suggested a deep learning-based method for detecting drones and classifying drone kinds. The algorithm is divided into two parts: detecting drones using a YOLO v2 model, and classifying drone types using convolutional neural networks (CNN) (Tricopter or quadcopter). The trained YOLO v2 achieved an accuracy of nearly 80% for both detection and classification tasks. It's worth noting that the detection and classification of drones using deep learning algorithms is an active area of research. Other approaches have been proposed in recent years, such as using radio frequency compressed signals to detect and classify UAVs. Another study proposed a hybrid deep learning model that combines LSTM and Bi-GRU classifiers to provide enhanced object detection in drone imagery for search and rescue operations.

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